

First launched into space on April 12, 1981, the Shuttle was a triumph of American technology and can-do spirit. Once described as a “space truck,” it proved a versatile spacecraft, carrying satellites into orbit, acting as a mini space station and laboratory, performing space repairs, and helping build and operate space stations. Two of the six Shuttles built were lost in accidents:

Challenger in January 1986 and *Columbia* in February 2003.

Atlantis made the final Shuttle flight on July 8, 2011.

Cargo bay measures
60 x 15ft (18.3 x
4.6m)

Wings used only when Shuttle
glided in to land – they had
no function in space

Elevon



OUT ON A LIMB

Astronaut Daniel Barry completes a space walk wearing a SAFER (simplified aid for EVA rescue). He is tethered to the Space Shuttle *Discovery* at his waist.

Three main engines
were supplied with
liquid hydrogen and
oxygen from external
tank during first eight
minutes of flight

Eleven video screens, five
computers, and over 1,000
switches made up the MEDS

Orbital maneuvering
system engines

Aft reaction control
system, used to
maneuver Shuttle into
correct position while
in space



SHUTTLE COCKPIT

This “fisheye” view of the *Atlantis* flight deck shows NASA’s Multifunction Electronic Display System (MEDS). This state-of-the-art system used technology found in many commercial airliners and reinforced Shuttle safety by providing multiple backup display functions.

Specifications (Atlantis)

LENGTH	122ft 2in (37.2m)
WINGSPAN	78ft 1in (23.8m)
HEIGHT	56ft 6in (17.3m)
ORBITAL SPEED	Mach 15+ or 18,000mph (28,800kph)
WEIGHT	2,000 tons (2,041 tonnes)
CREW	5-8

Thermal Protection
(TPS) tiles protected
the underside and other
surfaces of the Shuttle
from burning up on reentry
into the Earth’s atmosphere

THE ORBITER

From 1977, six Shuttle orbiters were built by NASA. *Enterprise* was used only for testing and never flew in space. *Columbia* was followed by *Challenger*, *Discovery*, *Atlantis* (shown here), and *Endeavour* (built to replace *Challenger* after its tragic loss in 1986).

DRAG CHUTE LANDING

The drag chute on the Space Shuttle *Endeavour* helped to slow it down on landing. The first use of a drag chute during a Shuttle landing was on *STS-49*, launched on May 7, 1992. The Shuttle descended to Earth as an unpowered glider, landing at a speed of around 219mph (350kph).

